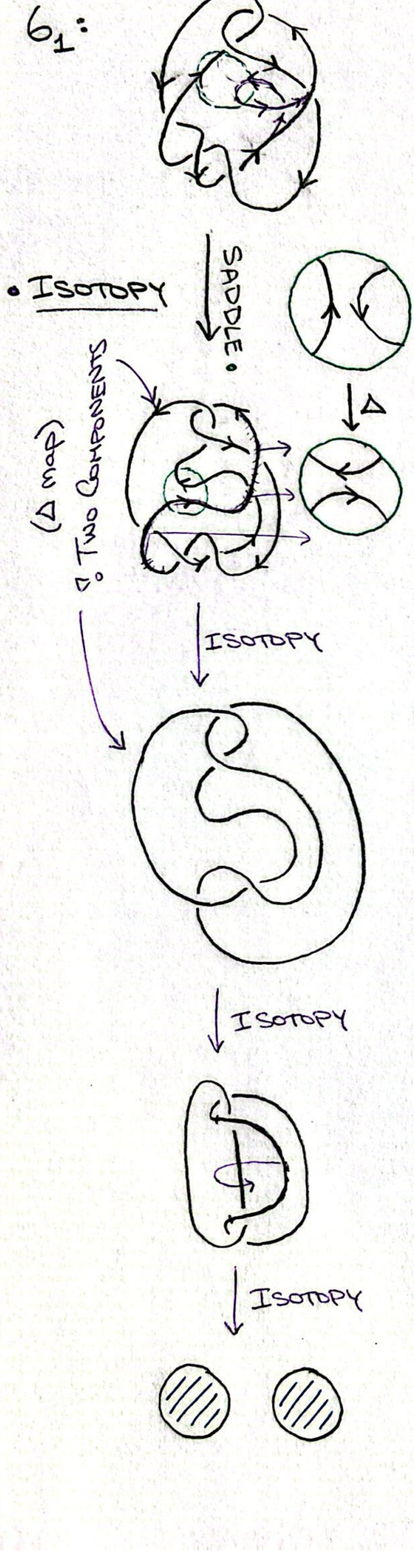
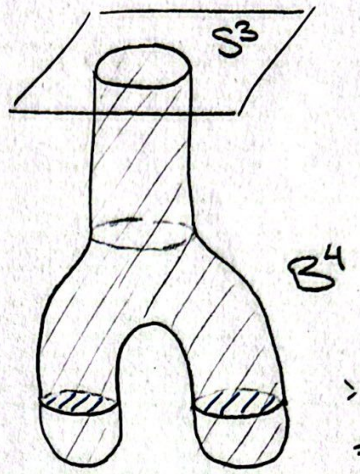
6_1:



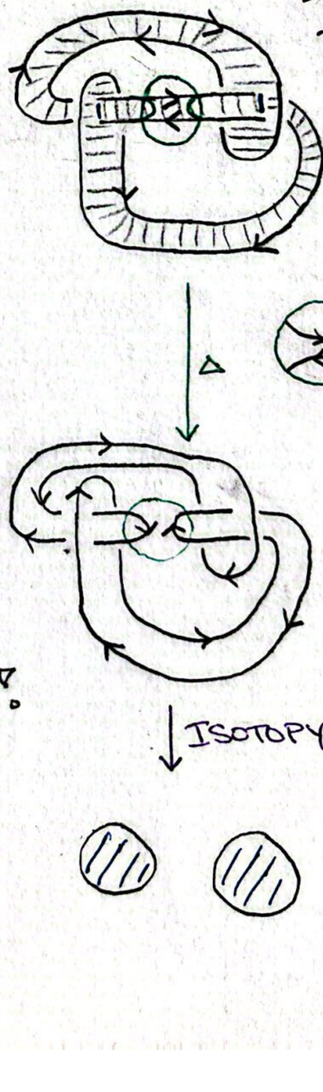
[LEMMA: RIBBON ⇒ SLICE]



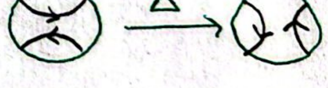
> SOLID DISK:
⇒ SLICE!

DEF: RIBBON: A knot is <u>ribbon</u> if it bounds a disk that crosses through itself only in arcs. In particular, $\boxed{K\#-K}$ is ribbon.

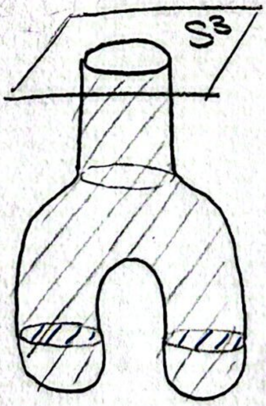
EXAMPLE:



> ARG: "PUSH INTO B4"
↳ not convincing? can also see via a saddle!



[Another way to represent 6_1]

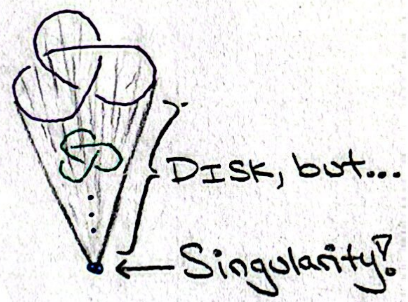


Again, a solid disk!
⇒ SLICE!

DEF: SLICE: $K \subset S^3$ is slice $\Longleftrightarrow$ K bounds a SMOOTH disk in B^4

↳ need SMOOTH, else note every knot bounds a cone in B^4, where pt is a singularity

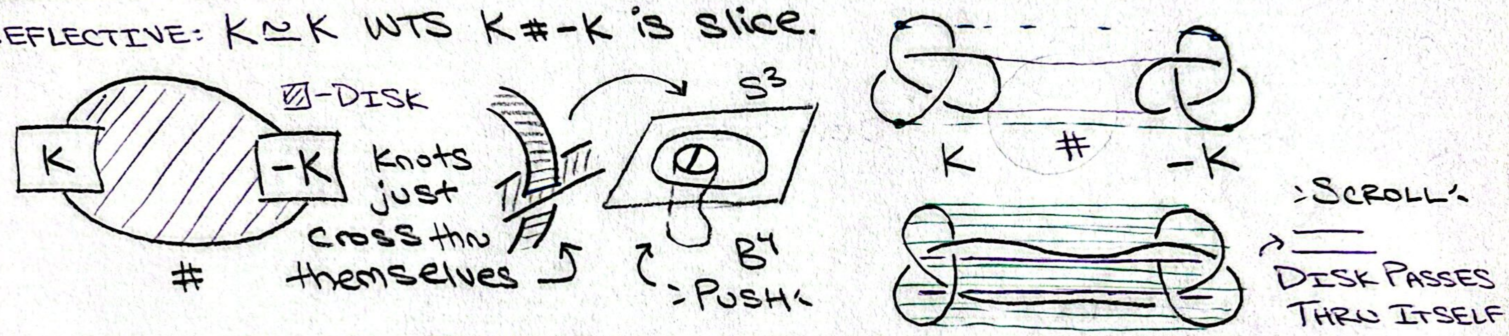
> think: how in calc smooth ⇒ nice, infin. diff. <



WHY SLICE IMPORTANT? > EQUIVALENCE RELATION! <

DEF: CONCORDANT: Knots K, J are concordant ($K \simeq J$) if $K \# -J$ is slice! Claim: Concordance is an equiv. rel. on knots!

REFLECTIVE: $K \simeq K$ WTS $K \# -K$ is slice.



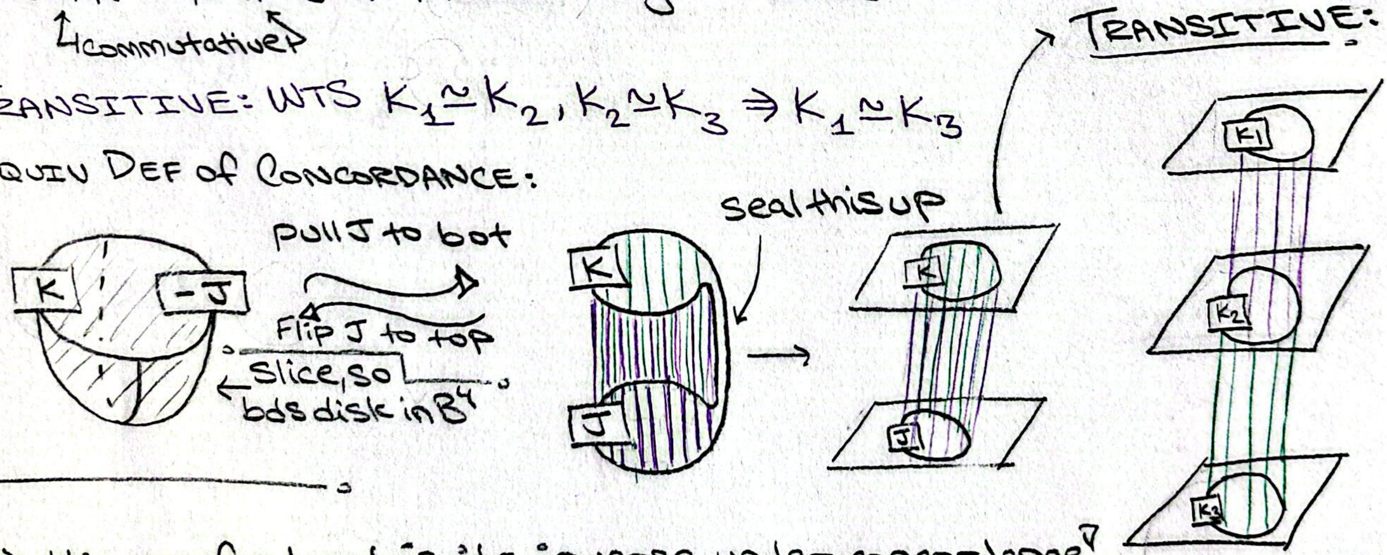
SYMMETRIC: $K \simeq J \Rightarrow J \simeq K$ Assume $K \# -J$ slice. WTS $-J \# K$ slice.

$-J \# K = K \# -J$ ← indeed is given slice

↳ commutative

TRANSITIVE: WTS $K_1 \simeq K_2$, $K_2 \simeq K_3 \Rightarrow K_1 \simeq K_3$

EQUIV DEF of CONCORDANCE:

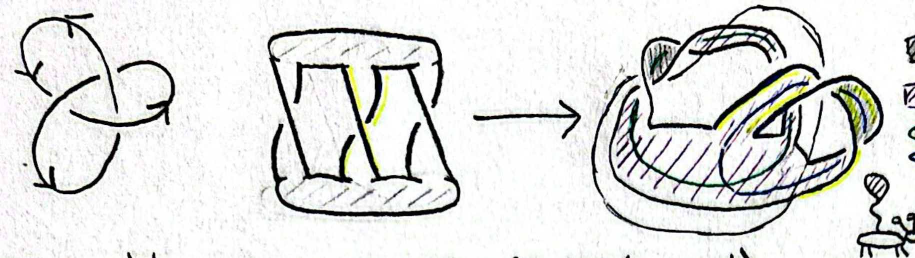


⇒ Mirror of a knot is its inverse under concordance!

$\mathcal{C}$ is the KNOT CONCORDANCE GROUP!

SEIFERT SURFACES Any knot bounds <u>orientable surface of disks & bands</u>

Seifert surface



SEIFERT MATRIX: each entry is $\ell k(x_i, x_j^+)$

eigenvals along diag

$$\begin{array}{cc} & \begin{array}{cc} x_1 & x_2 \end{array} \\ \begin{array}{c} x_1^+ \\ x_2^+ \end{array} & V = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \end{array} \rightsquigarrow V+V^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \to \sigma(\text{trefoil}) = \underline{2}$$

→ symmetric $V+V^T \Rightarrow$ diagonalizable

SIGNATURE OF MATRIX: #pos eigenvals − #neg eigenvals

$K \rightsquigarrow$ Seifert Surface $\rightsquigarrow$ Seifert Matrix $\rightsquigarrow \sigma(K)$ signature

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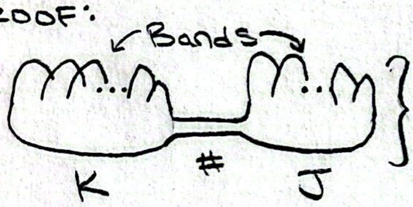
PROPERTIES OF SIGNATURE:

$\sigma(K\#J) = \sigma(K) + \sigma(J)$

PROOF:



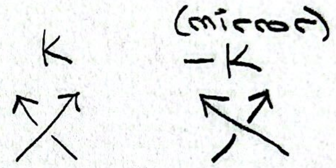
K loop lifts ⊔ from J loop lifts

$$V_{K\#J} = \begin{pmatrix} V_K & 0 \\ 0 & V_J \end{pmatrix} \Rightarrow V+V^T = \begin{pmatrix} V_K+V_K^T & 0 \\ 0 & V_J+V_J^T \end{pmatrix}$$

$$\Rightarrow D_{K\#J} = \begin{pmatrix} D_K & 0 \\ 0 & D_J \end{pmatrix}$$ diagonal breaks down

$\Rightarrow \sigma(K\#J) = \sigma(K) + \sigma(J)$. ▨

$\sigma(K) = -\sigma(-K)$



↳ all crossings flipped ⇒ linking #'s all flip

$\Rightarrow V_{-K} = -V_K \Rightarrow D_{-K} = -D_K \Rightarrow \sigma(K) = -\sigma(-K)$. ▨

K SLICE ⇒ B^4

Note, in B^4 so ▨, take tub. nbd: $F \times \underline{\underline{D^2}}$ so vert. circs bound D^2 (*)

By (*) we use thanos topology ⇒ the small vertical circles can shrink down and disappear!

$$\Rightarrow \exists V_K = \begin{pmatrix} 0 & A \\ B & C \end{pmatrix} \begin{array}{l} \}g \\ \}g \end{array}, \quad \underbrace{\quad}_{g}\ \underbrace{\quad}_{g}$$ where g is genus(K)

$\Delta[\cdot] \Rightarrow \approx \begin{pmatrix} 0 & D \\ D^T & 0 \end{pmatrix}$

$$\Rightarrow V_K + V_K^T = \begin{pmatrix} 0 & A+B^T \\ A^T+B & C+C^T \end{pmatrix}$$ $\Rightarrow g \times 1$ eigenvals, $\Rightarrow g \times (-1)$ eigenvals $\Big\} \sigma = \emptyset$

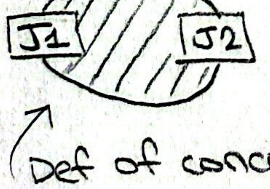
LEMMA: K Slice ⇒ $\sigma(K) = 0$ → PROOF

COR: Suppose J has Seifert surfaces F_1, F_2. Then $\sigma_{F_1}(J) = \sigma_{F_2}(J)$: well-def ∴ PROOF: $\sigma_{F_1}(J) - \sigma_{F_2}(J) = \sigma_{F_1}(J) + \sigma_{F_2}(-J)$

$= \sigma_{F_1 \# F_2}(J \# -J) = 0 \Rightarrow \sigma_{F_1}(J) = \sigma_{F_2}(J)$

COR: Suppose J1 concordant to J2.

$\Rightarrow \sigma(J1) = \sigma(J2)$. PROOF:



Def of concord.

$\Longleftrightarrow$ J1 # −J2 slice $\Rightarrow 0 = \sigma(J1 \# -J2)$

$= \sigma(J1) + \sigma(-J2)$

$= \sigma(J1) - \sigma(J2)$

$\Rightarrow \sigma(J1) = \sigma(J2)$. ▨

NOTE: converse, ie, $\sigma(K) = 0 \Rightarrow$ K slice IS NOT TRUE! Ex. (4,1), CONWAY K

Alexander poly: $V^T - tV$. So here, we specialize with $\underline{t = -1}$

$\Delta(-1)$ spec. case